

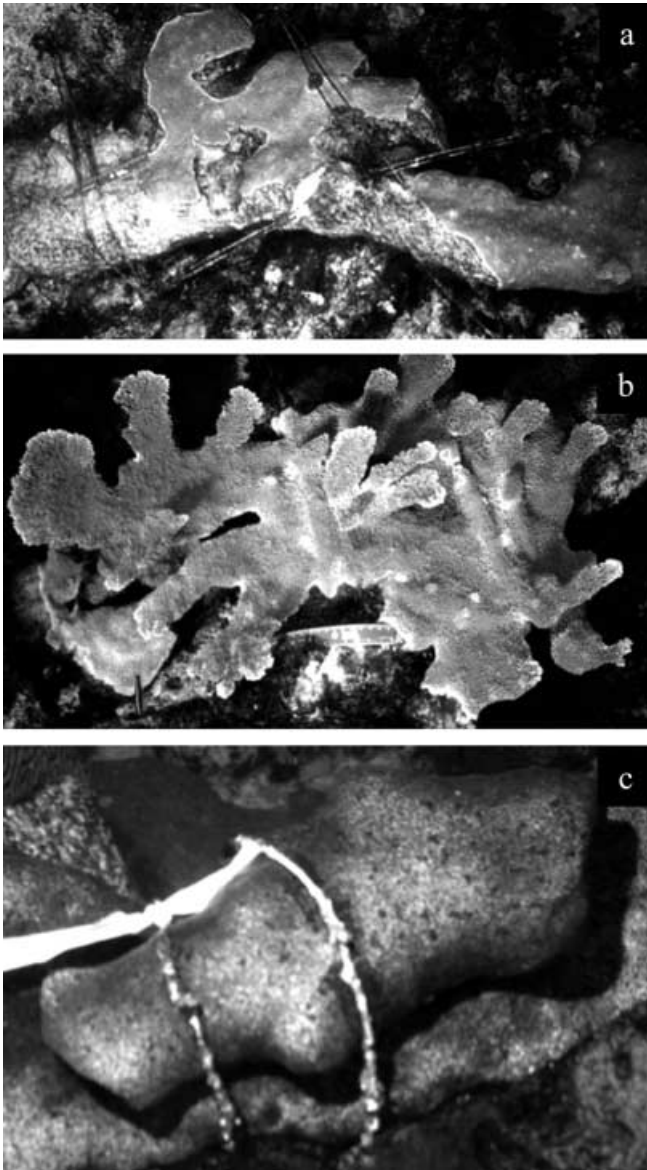


Fig. 2a–c Examples of restored *Acropora palmata* fragments at the Fortuna Reefer coral restoration site photographed in August 1999, approximately 2 years after restoration. **a** Fragment attached to the substrate, illustrating wiring technique used to stabilize it. Partial mortality occurred where tissue was in contact with the wire. Fragment is 104 cm in length. **b** Fragment reattached to an erect *A. palmata* colony using cable ties. This fragment exhibited fusion, growth over cable ties and well-developed proto-branches. Fragment is 82 cm in length. **c** Fragment attached to standing elkhorn skeleton that was colonized by *Cliona* species complex; fragment was subsequently overgrown by the sponge. Fragment is 26 cm in length

tissue (mean = 54%; 95% confidence interval = 50–58%) than those attached upside down (mean = 47%; 95% confidence interval = 40–52%). There was also a significant difference in fragment survival between the two attachment locations (chi-square analysis of a 2x2 contingency table, $\chi^2 = 6.2$, $df = 1$, $p = 0.013$), with a higher proportion of the living fragments found attached to the reef substrate. However, there was no difference in the

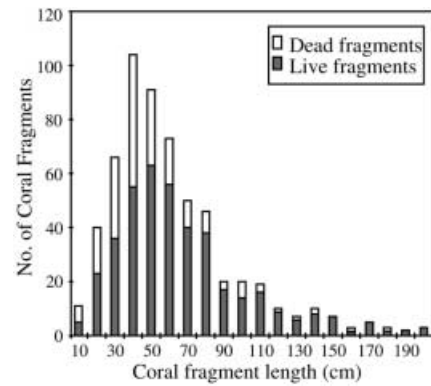


Fig. 3 Size frequency distribution of restored *Acropora palmata* fragments at the Fortuna Reefer coral restoration site. Number of fragments in each size class are shown as stacked bars

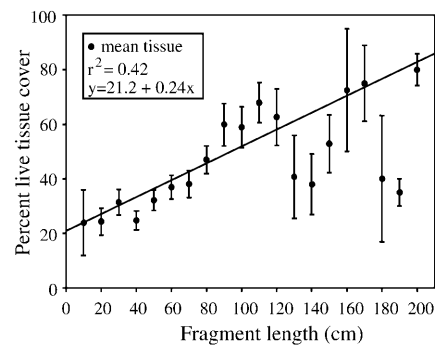


Fig. 4 Mean percent of live tissue ($\pm$ SE) on upper branch surfaces for surviving *Acropora palmata* fragments at the Fortuna Reefer coral restoration site. Data are pooled into twenty 10-cm size classes

amount of tissue remaining on live fragments with respect to attachment location (mean = 52%; $t = 0.47$, $df = 401$, $p = 0.64$; Table 1).

Effect of depth on fragment retention

Restored fragments were located from 2–7 m depth and were pooled into shallow (2–3 m), intermediate (3–4 m) and deep (>4 m or deeper) categories (Table 1). The greatest number of lost fragments were identified in shallow water (25% of the total sample examined in shallow water), followed by those in deep water (16%); fewer fragments from intermediate depths were missing (11%). The number of fragments missing also differed between attachment locations, with 84% of all missing fragments originally attached to the substrate. This may be an artifact of the data, however, as the tallied number of fragments attached to standing colonies and subsequently lost may be an underestimate. Since nails were not used when securing fragments to elkhorn skeletons, the only indication that a fragment previously existed there was the presence of remnant wire.

Effect of depth and location of attachment on fragment condition

The condition of remaining fragments, measured as the percent of surviving fragments and the amount of live tissue, was compared among depths and attachment locations. The length frequency distributions from the three depths did not differ significantly (pooled orientation and attachment location; single-factor ANOVA, $F=1.37$, $df=589$, $p=0.25$). Differences in survivorship were noted among depths (chi-square analysis of a 2×3 contingency table, $\chi^2=11.76$, $df=2$, $p=0.003$). More than 70% of the remaining fragments from shallow and intermediate depths survived, while only 59% of the deep fragments had survived. In addition, differences in the number of fragments that died were noted among depths for fragments attached to standing elkhorn skeletons (chi-square test, $\chi^2=9.26$, $df=2$, $p=0.01$), but not for fragments attached to the reef substrate (chi-square test, $\chi^2=3.33$, $df=2$, $p=0.19$). More than 50% of the fragments reattached to skeletons in deep water had died (Table 1). A two-factor ANOVA comparing the condition of the remaining live fragments with depth and attachment location indicates that no differences exist among depths for the percentage of tissue loss ($F=1.81$, $df=2$, $p=0.165$; Table 2). In addition, the amount of partial mortality on surviving fragments was not significantly different between attachment locations ($F=0.94$, $df=1$, $p=0.333$). In contrast, there was a significant interaction between attachment location and depth ($F=4.94$, $df=2$, $p=0.008$), with the greatest amount of tissue loss (62%) occurring in fragments attached to standing skeletons in deep water.

Discussion

Primary restoration at the *Fortuna Reefer* site was undertaken within 2 months of the grounding incident under the assumption that without rapid reattachment, a high percentage of fragments might die due to sand scouring and other factors, or be removed from the site during periods of high wave action (NOAA 1997a). To maximize the potential for reef recovery, the aim was to restore injured corals to the greatest extent possible. Few unrestored fragments (<2%) were tagged for controls (i.e., fragments created by the incident that were not reattached), because settlement monies from ship groundings and oil and hazardous materials releases may not be used to design and test hypotheses. The absence of data on the survivorship of control fragments provides an incomplete picture of the benefits gained by the considerable investment to secure the fragments. Nevertheless, a broad-scale census of the status of the wired fragments, as gauged by the extent of survivorship and regrowth, provides valuable information on the relative utility of the technique used to secure fragments, the relationships among survivorship and size, orientation, depth, and substrate, as well as information on

Table 2 Two-factor ANOVA examining the condition of surviving fragments from three depths attached either to relict reef substrates or to standing elkhorn skeletons. Condition was determined from amount of tissue remaining, using arcsine transformed percentages. *SS* Sum of squares; *MS* mean square

Source of variation	df	SS	MS	F	p
Depth	2	2,236	1,118	1.81	0.165
Attachment location	1	580	580	0.94	0.333
Interaction	2	6,095	3,047	4.94	0.008
Error	399	243,144			
Total	404	252,055			

potential sources of mortality. This information can be used to enhance future restoration efforts for this coral species and may assist in determining the need for subsequent corrective actions to enhance fragment survivorship.

To become successful recruits, *A. palmata* fragments must fuse to the substrate to prevent removal by wave action. The wiring technique used to stabilize fragments at the *Fortuna Reefer* grounding site was developed with the intent to enhance natural processes of fragment stabilization. It was used as an alternative to cement or epoxy due to the high wave exposure typical at this site and the amount of time required for these materials to harden. In August 1999, both living and dead fragments that were securely attached with wire were observed throughout the restoration site. Broken wire was also common; wire had loosened on some fragments and others had broken free and were carried away. Overall, 17% of the fragments that were restored originally were identified as missing. Fragment loss was greatest in shallow water, as evidenced by the number of nails, and/or wire without associated fragments. It is likely that more fragments were removed that could not be identified, however. Nails were not used to secure fragments to *A. palmata* skeletons and fragments and wire may have been removed completely, rendering the reattachment site undetectable.

Although wire failure appears to be a significant cause of transplant loss, these losses were relatively low considering that this reef is routinely exposed to severe weather and the site was directly impacted by Hurricane Georges in 1998. Studies that have followed the natural fate of hurricane-generated fragments have reported large losses in the months following the disturbance. In Florida, during an 11-month period in which no additional storms occurred, 57% of the fragments were removed from a hard-bottom substrate (Lirman and Fong 1997b). Similarly, over a 4-year period in Puerto Rico, a tropical storm and two hurricanes removed greater than 96% of *A. palmata* fragments that were generated on an offshore patch reef by a previous storm (Bruckner, unpublished data).

In many cases, the wiring technique appeared to hamper the growth of coral tissue, as evidenced by the absence of tissue directly in contact with the wire. Wire often separated two or more patches of tissue on frag-

ments that otherwise exhibited little or no tissue mortality away from the wire. The wiring technique did not inhibit tissue growth on all *A. palmata* fragments, however, since 16% of the fragments had tissue growing over the wire. Tissue mortality associated with the wire might have been reduced by using plastic-coated stainless-steel wire.

Previous observations of the restoration site indicated that the wire was colonized by filamentous algae and macroalgae and algae were killing tissue in places (E. Weil, personal communication, April 1998); this was less common 3 months later, however (M. Miller, personal communication). During the present survey, algae (primarily filamentous green algae, *Dictyota* spp., and cyanobacterial mats) were observed on the wire associated with a relatively small proportion of the living fragments (3%). The presence of algae did not appear to be selectively affecting the wired fragments during this study, since dense macroalgae also occurred away from the wire and along the margins of unrestored *A. palmata* fragments.

The extent of survival and potential for regrowth may be affected by fragment size. Modular organisms are known to allocate different proportions of their energy and resources to reproduction, with larger colonies of some species investing more energy in reproduction than in growth (Jackson and Hughes 1985). If this is the case for *A. palmata*, smaller fragments potentially have a better chance of survival because they may concentrate their resources on growth, increasing their ability to reattach rapidly. The effect of size on survivorship of *A. palmata* fragments has been studied previously with conflicting results, however. Highsmith et al. (1980) observed a positive correlation between fragment size and survivorship, while Rogers et al. (1982) noted that fragments that survived were smaller overall than those that died. Lirman and Fong (1997b) did not observe any relationship between fragment size and survivorship, possibly because the majority of the fragments were small. In the present study, the mean size of fragments that died was significantly smaller than that of the live fragments, with the highest rate of mortality observed among fragments under 50 cm in length.

Reef-building corals are affected by partial colony mortality due to various factors. Tunnicliffe (1981) noted that Caribbean acroporids exhibit the fastest growth at the distal edges of the colonies and older, basal regions sustained higher rates of mortality. Lirman and Fong (1997b) observed tissue loss in many (73%) of their tagged fragments; fragments lost 61% of their original tissue area over an 11-month period, but surviving fragments had begun to develop vertical growth features. In the present study, surviving fragments had tissue on approximately half of their upper branch surfaces after 2 years. Although reattached fragments on the *Fortuna Reefer* site appear to have fared better than unrestored fragments in Florida and elsewhere in Puerto Rico, differences may be related to specific environmental parameters that vary among locations. In addition,

the amount of tissue remaining on the branch surface may be less important for long-term survivorship than the ability of fragments to reattach and develop new vertical branches. In both studies, fragments had lost a considerable portion of their living tissue, but many were developing proto-branches, indicating that continued growth was occurring.

A critical factor affecting the ability of a fragment to successfully recruit is the substrate type where the fragment is deposited (Lirman and Fong 1997b). Bak and Criens (1981) found that *A. palmata* fragments fuse to other, living elkhorn fragments faster than to any other type of substrate. In Florida, elkhorn fragments landing on rubble substrate had a higher survivorship and a lower percent tissue loss than those deposited on relict reef substrate (Lirman and Fong 1997b). At the *Fortuna Reefer* grounding site, fragments were secured to the relict reef substrate or to dead, standing elkhorn skeletons. Those fragments attached to skeletons were expected to exhibit enhanced survivorship associated with a reduction in scouring, improved water circulation, increased light exposure, and possibly a reduced exposure to benthic predators and pathogens. Contrary to this hypothesis, fragments secured to the reef substrate had a higher survivorship than those attached to elkhorn skeletons. Although fragments on the reef substrate were larger, the percentage of tissue remaining after 2 years on survivors did not differ between the two attachment locations. The differences in survivorship are likely to have resulted from a higher rate of mortality experienced by those fragments placed in contact with the invasive sponge *Cliona* spp., which was found commonly overgrowing elkhorn skeletons and less frequently on the substrate.

It is important to note that elkhorn populations have undergone a regional decline over the last two decades (Gladfelter 1982; Bythell and Sheppard 1993). Large-scale catastrophic losses have been associated with physical disturbance from hurricanes, but a widespread, white-band disease (WBD) epizootic is a contributing factor that has certainly accelerated the decline and continues to afflict elkhorn colonies throughout the region (Gladfelter 1991; Aronson and Precht 2001). Chronic partial mortality from natural stressors like WBD, predation, and other factors not associated with the restoration may be largely responsible for the inability of fragments to successfully fuse to their attachment sites. For instance, the undersides of branches on intact colonies such as those previously affected by WBD often lack live tissue. Should they become fragmented, they may be unable to reattach without tissue in direct contact with the substrate (Bruckner, unpublished data).

In this study, recent signs of mortality including WBD were observed on a small proportion of the restored fragments, as well as on several surrounding *A. palmata* colonies that were unaffected by the ship grounding. Given a widespread occurrence and a presumed increase in the prevalence of coral disease,

predators, and other stressors affecting acroporids in Puerto Rico (Bruckner et al. 1997; Williams et al. 1999), it is unlikely that mortality of restored fragments is exclusively related to the restoration. It is possible that stabilization of *A. palmata* fragments may be an activity that has only slowed an inevitable monotonic mortality process. Nonetheless, studies in Puerto Rico and Florida have shown that unrestored fragments continue to grow, but even sizeable fragments in good condition are lost during periods of high wave energy if they do not have a secure connection to the substrate (Lirman and Fong 1997b; Bruckner, unpublished data). Results to date from the *Fortuna Reefer* site emphasize the importance of a timely restoration. Without intervention, this site might have mirrored other locations in Puerto Rico, where almost complete removal of fragments occurred during subsequent storms due to low levels of natural fusion (Bruckner, unpublished data).

Adaptive management may be particularly valuable as this coral species has been declining throughout its range since the late 1970s, and innovative techniques that enhance survivorship may be critical in promoting localized recovery of elkhorn populations. In anticipation of the continued degradation of the stainless-steel wire which begins to corrode at points of crimping and twisting and given the large number of fragments that were unable to fuse to the substrate, corrective action to maintain the remaining fragments on the site may increase the overall effectiveness of initial restoration efforts. Recommendations for altering the original technique include the use of Portland cement or two-part marine epoxy (Hudson and Diaz 1988; Gittings et al. 1994), in conjunction with heavy-gauge monofilament line, nylon or Teflon-coated wire, or plastic cable ties for short-term fragment stabilization while the adhesive sets. It is imperative that those fragments exhibiting high rates of growth and fusion should not be included in corrective restoration efforts. Corrective action for the remainder will continue to maintain restored fragments on site, providing additional time for recovery and enhancing the long-term success of the initial restoration. This technique would benefit from the coordinated efforts of a multi-task team to remove the wire from loose fragments, scrape and clean the undersides of fragments and the reattachment sites to remove algae and other encrusting organisms, apply the cement or epoxy, and temporarily secure the fragments to allow the adhesive to set. Furthermore, a portion of the fragments restored in 1997 should be left unmanipulated in order to allow an assessment of the benefits gained by a mid-course correction and the potential value of future restoration efforts.

Restoration is a relatively new science that soon may need to be applied on a larger scale to mitigate effects of hurricanes, bleaching events, and disease outbreaks (Precht 1998). Current research is focusing on methods to enhance coral recruitment, restore the natural balance of herbivores and maintain coral nurseries to rescue and rehabilitate fragments generated by physical distur-

bances, for use as transplants for degraded areas. The application of new technologies like those used in the *Fortuna Reefer* restoration can continue to advance the science. Emphasis must be placed on scientifically based approaches needed to assess the outcome of restoration efforts and improve techniques to restore, rehabilitate, and replace damaged corals.

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